Stark Shifts in Simulated Excited State Vibrational Spectra

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Abstract

Our chosen system of study is the *2-[(7-4-[N,N-Bis(4-methylphenyl)amino]phenyl-2,1,3- benzothiadiazol-4-yl)methylene]propanedinitrile* (DTDCPB) chromophore prevalent in organic photovoltaic materials research. DTDCPB has nitrile groups that make it ideal for IR studies due to its strong signature stretching frequency. For the first component of our project, we hope to simulate some ground state IR spectra of DTDCPB in the presence of varying external fields. This can then be used to calculate the Stark shift and build a calibration curve for the effect of external field on the Stark shift of this molecule. Furthermore, we plan to probe the excited state of DTDCPB as well and see if we can look at Stark effects in the excited state as well. The third step we hope to investigate is a pump-probe transient IR study of this molecule where the excited state dynamics of DTDCPB can be observed. Hopefully our understanding of Stark effects in the ground and excited states will elucidate some of the dynamics of the transient IR study. **CURRENTLY 175 WORDS**

Introduction

As research in solar energy becomes increasingly important for environmental applications, several new strategies are being investigated to increase the efficiency of these devices. Specificlly in organic photovoltaics (OPVs), one such strategy has been to use small molecules with large ground-state dipole moments. It is believed that these dipoles can allow for enhanced hole hopping and can minimize charge recombination (ref Griffith). One particularly promising molecule is 2-[(7-4-[N,N-Bis(4-methylphenyl)amino]phenyl-2,1,3-benzothiadiazol-4-yl)methylene] propanedinitrile (DTDCPB) and has resulted in OPVs with power conversion effiencies of up to 9.6%. The nitrile groups present in the molecule can act as reporter moieties for vibrational spectroscopy. By modeling the effect of an applied electric field on absorption spectrum (Stark Effect), it is then possible to gather information regarding the local electric field during charge transfer events in a photovoltaic device via excited state IR spectroscopy. The first requires a thorough understanding of the electric field dependence of the ground state IR spectrum before probing the excited state.

From a rational materials development standpoint, the ability to understand the mechanisms behind these phenomena enables the tuning and enhancement of current materials in order to create novel, high-efficiency semiconductor and photovoltaic devices. Therefore, computational modeling of these systems is a powerful tool in our kit for developing advanced materials. This study was carried out in several stages. First, the equilibrium properties of the DTDCPB molecule for the ground state were obtained through frequency calculations in *Gaussian 09*. The Stark effect was modeled and calibrated by applying external static electric fields of various magnitudes to the optimized equilibrium geometries and performing the same frequency calculations as in the model without an applied field. The changes in frequency were obtained from the frequency calculations and compared to calibrate the Stark shifts for the various field strengths. Subsequently, Born-Oppenheimer Molecular Dynamics were used to obtain model spectra using a different set of methods from the standard frequency calculations in *Gaussian 09*.

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Theory

Standard Frequency Calculations

First, a brief description of how the *ab initio* frequency calculations are performed in the **g09** distribution of Gaussian is in order. The *normal modes of vibration* are calculated by diagonalizing a matrix of the second derivatives of the electronic potential energy surface (PES) for the system of N atoms. The eigenvalues of this matrix, called the *Hessian*, correspond to the frequencies of the normal modes of vibration for the system. More explicitly, this matrix is obtained from the Taylor expansion about the potential energy minimum with respect to nuclear position q_k. This expansion varied by the gradient g_k is given by Equation 1.

$$V(g_k) = V(0) + \sum_k \left(\frac{\partial V}{\partial q_k}\right) q_k + \frac{1}{2}\sum_{j,k} q_j \left(\frac{\partial^2 V}{\partial q_j \partial q_k}\right) q_k + \frac{1}{6}\sum_{j,k} q_j \left(\frac{\partial^3 V}{\partial q_j \partial q_k \partial q_l}\right) q_k q_l + \cdots \quad (1)$$

where q_k corresponds to the potential energy surface coordinate, g_k is the gradient, and $j, k = 1, \cdots, N$. We find that the Hessian is our second geometric derivative term $H_{j,k} = (\partial^2 E/\partial q_j \partial q_k)$ in the Taylor expansion, and we can more compactly rewrite Equation 1 as

$$V(g_k) = V(0) + \sum_k g_k + \frac{1}{2}\sum_{j,k} q_j H_{j,k} q_k + \frac{1}{6}\sum_{j,k} q_j A_{j,k,l} q_k q_l + \cdots \quad (2)$$

Where the gradient g_k is defined by the first geometric derivative, the Hessian matrix element is given as $H_{j,k}$ and the anharmonicity tensor element A_{jkl} is defined by the third geometric derivative with respect to q_k in Equation 2. The Hessian and anharmonicity tensors are of dimensions $N \times N$ and $N \times N \times N$ respectively. Diagonalizing the Hessian to obtain the set of N eigenvalues $\{\omega_k\}$ results in the orthogonal set of normal mode frequencies in the frequency domain. To obtain the intensities of the transitions, we use the fact that the vibrational intensities are proportional to the square of the first geometric derivative of the *dipole moment* (Equation 3), which is itself derived from the instantaneous charge density around the nuclei in the molecule.

$$\mathcal{A}_{\tilde{\nu}}(q_k) \propto \left(\frac{\partial \mu}{\partial q_k^2}\right) \tag{3}$$

Scaling the frequencies $\{\omega_k\}$ by the intensity $A_{\tilde{\nu}}(q_k)$ gives us a simulated vibrational spectrum for the molecule from purely *ab initio* calculations.

The Stark Effect

The Stark effect is a quantum mechanical phenomenon in which the energy level splittings are increased in the presence of an electric field. The effect is readily measured via changes to the vibrational and optical spectra of a species **(Figure?)**. We can also model it computationally by performing electronic structure calculations accounting for the presence of an external field. The observable effect is that of a change in dipole moment ($\Delta\mu$) which is dependent on the degree of charge separation for a transition. Additionally, we observe a change in polarizability ($\Delta\alpha$) which corresponds to the change in sensitivity of the vibrational transition to the applied field. The change in frequencies is modeled in Equation 4 where $V(r_k)$ is the potential with respect to position r_k of electron k around the nuclei.

$$\Delta\omega_k = -\Delta\vec{\mu} \cdot \left(\frac{\partial V(\vec{r})}{\partial \vec{r}_k}\right) - \frac{1}{2}\Delta\alpha \cdot \left|\frac{\partial V(\vec{r})}{\partial \vec{r}_k}\right|^2 \tag{4}$$

Born-Oppenheimer MD (BOMD)

This work makes use of an *ab initio molecular dynamics* technique called Born-Oppenheimer Molecular Dynamics (BOMD) to derive the model spectra for DTDCPB both with and without the static electric field present. This technique uses Born-Oppenheimer approximate time-*independent* density functional theory (DFT) calculations to compute the electronic potential at each nuclear evolution time step, yet uses classical equations of motion (EOMs) integrated with the Velocity Verlet integration method (Equations 5) to approximate the nuclear dynamics as a time evolution.

$$\begin{aligned} \vec{x}(t+\Delta t) &= \vec{x}(t) + \vec{v}(t)\Delta t + \frac{1}{2}\vec{a}(t)\Delta t^2 \\ \vec{v}(t+\Delta t) &= \vec{v}(t) + \frac{\vec{a}(t)+\vec{a}(t+\Delta t)}{2}\Delta t. \end{aligned} \tag{5}$$

In this context, BOMD has distinct advantages over the standard calculation technique in *Gaussian* of diagonalizing the Hessian as discussed above. The Stark effect is most readily observed in vibrational spectra, thus using purely *ab initio* techniques we require either the second or third geometric derivative of the electronic PES, which can be computationally prohibitively expensive for larger systems. In contrast, BOMD only requires the gradient of the electronic PES to compute the EOMs for the classical nuclear dynamics (Equations 5) and only the electron density is needed to compute the time-dependent dipole moment (Equation 6), the time series from which the model spectra are derived. In this manner, BOMD manages to capture a useful amount of the anharmonicity and higher order terms in the Taylor expansion of the potential given by Equations 1 and 2 without having to explicitly calculate anything more expensive than the gradient for each time step.

$$\vec{\mu}(t) = -\sum_{mu}\sum_{nu} P_{\mu\nu}\big(\nu|\mathbf{r}(t)|\mu\big) + \sum_{A} Z_A \mathbf{R}_A(t) \tag{6}$$

Obtaining Spectra from Dipole Orientation Time Series

Since BOMD only calculates the first geometric derivative of the PES, we need to obtain the vibrational spectra another way than the normal path followed by a frequency calculation in *Gaussian*. Since the dipole vector $\vec{\mu}(t)$ for each time t is output from BOMD as a time series, we can take the first time derivative $\dot{\vec{\mu}}(t)$ from the time series and use the *dipole autocorrelation function* $\langle\dot{\vec{\mu}}(t), \dot{\vec{\mu}}(t+\tau)\rangle_\tau$ as the kernel of an integral transform resembling a Fourier Transform (Equation 7 to obtain the normal mode frequencies and intensities.

$$\mathcal{A}(\omega) \propto \int \langle\dot{\vec{\mu}}(t), \dot{\vec{\mu}}(t+\tau)\rangle_\tau e^{-i\omega t} dt. \tag{7}$$

Plotting the function $\mathcal{A}(\omega)$ along the frequency domain with arbitrary intensity values yields a model spectra very similar to that obtained from diagonalizing the Hessian in an *ab initio* calculation.

Methodology

The Stark effect comes in when a field is applied.

Why are we using a static field?

First the molecular geometry was optimized in *Gaussian* at the *B3LYP/6-311G(d,p)* level of theory both with and without the presence of a static electric field generated by a dipole oriented along the z-axis perpendicular to the plane within which most of the DTDCPB molecule is embedded (figure?). The nuclear dynamics calculations needed to simulate vibrational spectra were carried out in various dipole field strengths defined in *Gaussian* in order to "calibrate" the Stark shifts in the product spectra. Then, thermal sampling is carried out in the same electric field to generate trajectories for nuclear dynamics in a separate job using the optimized geometry.

Easy to put in an external field (molecule comes into Equilibrium in the field) to see the IR

The molecule is geometry optimized, then frequency calculated

Static = x,y,z then value of the field

Geometry optimization in a static E field (google the warnings to explain it)

Include point charges (dummy charges)

Dipole along z-axis, put two point charges on the axis and vary the distance to vary magnitude of the field. Z-axis is normal to the plane of the molecule

In order to efficiently obtain vibrational spectra of the full DTDCPB dimer system and model the experiment BOMD was first applied to a cyanide ion as a test case to coarsely model the nitriles on DTDCPB. Subsequently, it was applied to DTDCPB monomer, and finally to the DTDCPB dimer system. BOMD is an *ab initio* molecular dynamics method which couples quantum mechanical calculation of the electronic structure and potentials to a classical treatment of nuclear dynamics. We chose to use BOMD for the larger molecular systems specifically because it is able to capture more of the physics of the system of interest more efficiently than the traditional method of calculating the Hessian to obtain the frequency eigenvalues.

Computational Results

Discussion

Conclusion

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